A SynBio Whodunnit!

The activity is best run with 4 people present. One person at each station and a coordinator to greet people, and give prizes at the end. Each of the activities are built to be done independently, giving each a piece of information at the end to be reported back to the coordinator. When someone arrives you should greet them, for example you could say:

“Hi, welcome to our murder mystery science camp. Phil Jasmine, a local philanthropist was recently killed in his mansion. The perpetrator slipped poison into his drink, and we have the suspects in custody. In order to solve the crime you must go to 3 different departments to figure out who killed Phil, where it happened, and what poison they used. When you have that figured out, come back to me for a prize at the end.”

The participants will then go to each station to conduct the experiments listed on the individual sheets.

Some notes on each activity:

- The strawberry DNA extraction encourages the participants to take the extract with them in a sealed ziplock bag.
- If any of the participants are queasy feel free to tell them that the sample is strawberry mash.
- The pH test strips should be stored in the box, make sure not to leave more than1 out at a time.
- The chromatography experiment uses dye, make sure to tell the participants to be careful to not spill on their hands or clothes.

Activity Instructions

DNA Extraction:

- Before the activity mash up 16oz of frozen strawberries and separate into individual cups
- Set out all of the supplies listed in the sheet for the activity.

- At the beginning of the activity explain the instructions written on the activity sheet
- Once the DNA is extracted, ask for a sample of the extracted DNA, place it into a separate box, and then reveal that the DNA matches Ted the cook
- At the end of the activity invite the participant to put their DNA extract in a ziplock bag to keep

pH Test strips:

- Before the activity, soak one cotton ball in vinegar, one in water, and one in water with baking soda dissolved
- Place the balls in cups with the following labels:
    - Bathroom: Water
    - Kitchen: baking soda
    - Dining room: vinegar
- During the activity outline the instructions on the sheet, and provide insight into what pH strips do, as outlined on the activity page.

Chromatography:

- Before the activity, pour 3 drops of red and green food coloring into a sample cup and mix with a few oz of water
- Be careful throughout the activity to prevent the ink from spilling, we recommend you keep the ink out of reach of the participant when it is not being used
- Place pipettes into a cup for use
- If you are nearby a water tap you do not need to keep a container of water, but we do recommend it
- Grab a trash container and keep it nearby to prevent ink spillage
- During the activity outline the instructions on the sheet, and provide insight into what pH strips do, as outlined on the activity page.
- After the activity, invite the participant to keep their chromatography paper in a plastic bag, but make sure to dispose of all materials that used ink carefully

DNA Extraction

The suspect left some samples at the scene of the crime. In order to figure out who did it, you must extract the DNA from the sample and bring it to a technician at the station to identify who's DNA it is. At the end you will be told which suspect did it, to submit to the coordinator at the end of all tables.

Here is a list of the suspects:

- Gerard the buttler
- Ted the cook
- Jannice the financial advisor
- Jen the doctor
- Ben the cleaner

Once given the sample you will find the following materials:

- A cup of sample
- Dish soap
- Salt
- Water
- Coffee filters
- Coffee stirrers
- Plastic spoons

In order to extract the DNA you must

1. Add half a spoonful of dish soap and half a spoonful of salt to the solution
2. Fill the cup halfway with water and stir well
3. Pour the new mixture through a coffee filter into a new cup
4. Add rubbing alcohol equal in volume to your filtered mixture
5. Wait until the DNA itself separates in a white layer on top
6. Use the coffee stirrers to take a sample of DNA and give it to the technician.

Congratulations! You just extracted DNA–which you can take home if you put it into a bag. You discovered who did it.

pH Testing Strips

In order to fully solve the case you must also know where the crime occured. The poison used was slightly acidic, and some should have spilled on the ground but the perpetrator covered their tracks. We have cotton balls that have been rubbed around various rooms of the crime scene. Identify which room contains the acidic poison.

You might be asking: What does acidic mean? Acidic compounds have a low pH, and basic compounds have a high pH. Everything is compared to water which has a pH of 7, everything lower than 7 is typically considered acidic, and everything above 7 is considered basic. We use pH strips that change color to tell what the pH of a solution is.

Take the pH strips at the table and tap the cotton balls against them to figure out what room the crime was done in. Once you have solved it, tell your answer to the person at the table to get a location card.

ID by Chromatography

The poison used was mixed into a drink, and because of that we cannot easily identify the poison. We need to separate out the toxin in order to compare it to known compounds.

Scientists often use a technique called chromatography, where we leave some mix of chemicals on a material, and soak it, the liquid separates out the compounds and these results can be compared to pure compounds.

In this experiment you will need to add a drop of the toxin given onto a paper towel, then soak one end of the towel in water. The water should separate out the components, and you can compare the color to other toxins with known colors using the cheat sheet below:

- Snake venom: yellow
- Poisonous berries: blue
- Spider venom: red
- Cyanide: green